County income and the visibility of plant based dining

Youmin Li | Dataset option 2

Background and motivation

Plant-based diets can reduce several environmental burdens of food consumption (Clark et al., 2019). The restaurants category and location data set may help make these choices visible, but a vegetarian tag does not measure the nutritional quality, price, or environmental footprint of a meal.

The dataset of US Restaurants & Food Outlets included categories and geo-coordinates information. Therefore, a basic empirical question precedes policy evaluation: where are explicitly plant-based dining choices visible, and how does the answer depend on their definition?

I will use a dataset of the U.S. restaurant and add income census data to analysis relationship between household income level and share of vegan restaurant in county level.

Research question

Among listed eating outlets in the United States, is higher county household income associated with a larger vegan or vegetarian share, conditional on population density and state fixed effects? The comparison of two outcome definitions tests whether that pattern reflects specialist establishments or the broader visibility of menu options through platform categories. Neither interpretation alone establishes actual consumption.

Data and processing

I used `380K_US_Restaurants.csv` from Kaggle version 2 (kwxdata, 2023). The data set included repeated location according to international entries. I retain explicit United States addresses with valid coordinates, resolve duplicates using location identifiers and a normalized name–coordinate fallback, require observed category fields, and apply a documented eating-outlet category rule. Among duplicate records, richer category information receives priority; ties use valid ratings, address completeness, and raw row order.

Full resolution Census TIGER polygons assign counties; unmatched or multiply matched points are excluded rather than snapped. Census SAIPE 2021 supplies median household income, and Vintage 2021 population estimates and TIGER land area supply population density (U.S. Census Bureau, 2021b, 2022a, 2022b). Five-character county FIPS codes link these sources with unique-key checks. The resulting 135,072 locations are reduced to 131,807 listings in 862 counties across 48 states by requiring at least 30 listings per county. Counties with zero tagged outlets remain in the sample. Appendix table 1 reports sequential attrition.

Empirical strategy

The specialist indicator requires a primary category of vegan restaurant, vegetarian restaurant, or vegetarian cafe and deli. The broader indicator also accepts either term in the secondary category list. Both use the same observed-category denominator. For each definition, I estimate

$$100\frac{V_c}{N_c} = \alpha_{s(c)} + \beta \log_2 M_c + \gamma \log_2 D_c + u_c, \qquad w_c = N_c,$$

where V_c is the tagged count, N_c is all retained listings, M_c is median income, and D_c is residents per square kilometer. Weighting by N_c targets a listed-outlet-weighted linear association; it does not produce population representative estimates. The coefficient β is percentage points per income doubling. HC1 state clustered covariance, a cluster count correction, and t_{G-1} intervals allow within-state dependence (Zeileis et al., 2020). Sensitivity checks change weights, the county threshold, outlet definitions, and functional form.

Key findings

The sample contains 3,603 broadly tagged outlets (2.73%) but only 365 primary category specialists (0.28%). Taco Bell accounts for 27.3% of broadly tagged outlets, illustrating how secondary labels and specialist status describe different market features. Such labels may reflect real menu options; their presence is not itself a coding error.

Appendix Figures 2 and 1 show the geographic distribution of county-level specialist shares and the unadjusted income gradients, respectively. Using U.S. Census cartographic boundaries (U.S. Census Bureau, 2021a), the map displays the share of listed eating outlets classified as vegan or vegetarian specialists. Among counties in the analysis sample, higher shares appear in parts of the Northeast, along the West Coast, and around Miami and Denver, while lower shares are observed across much of the Midwest and Rocky Mountain region. These descriptive patterns are broadly consistent with the hypothesized association between plant-based dining and county income, but do not establish causality. Local food cultures, consumer preferences, climate, and other unobserved factors may also contribute to the observed distribution.

Regression result shown in table 2: the conditional broad-tag coefficient is 0.561 percentage points per income doubling (95% interval [0.047, 1.075]). Moving between the county income quartiles, from $53,611 to $72,708, corresponds to a fitted linear contrast of 0.247 points [0.021, 0.472], holding the other regressors fixed. For primary specialists, the coefficient is 0.024 [-0.104, 0.152]. This imprecise estimate does not establish the absence of a specialist income gradient.

Excluding Taco Bell from both numerator and denominator increases the broad coefficient to 0.710 [0.137, 1.283]; that brand alone does not generate the positive association. The broad coefficient also remains positive when each state is omitted in turn, ranging from 0.470 to 0.724. However, equal county weighting produces an interval spanning zero. Poisson models with listing-count offsets give an income-doubling rate ratio of 1.23 [1.01, 1.49] for broad tags and 1.09 [0.69, 1.73] for specialists. Zero-total states are omitted only in the latter nonlinear fit, with its separate sample reported in the code output.

Discussion and implications

The evidence is consistent with a greater share of labeled plant-based options in higher-income counties. It provides substantially weaker evidence for a gradient in specialist establishments after density and state differences are taken into account. However. counting all vegetarian-tagged outlets as dedicated vegetarian restaurants would greatly overstate specialization. Therefor, the present association cannot show that increasing income, subsidizing an outlet, or adding a category label would change diets or emissions. Secondary tags may nevertheless be useful for studying consumer information and how generalist restaurants market plant-based options.

Limitations and future steps

The listing source is a convenience sample with undocumented collection dates, incomplete geographic coverage, and unknown inclusion probabilities. Using 2021 socioeconomic variables does not resolve reverse causality or omitted variables. County income may proxy education, tourism, urban form, or preferences; density and state fixed effects cannot isolate a demand mechanism. County associations also cannot identify individual income differences or within-county access inequalities.

Category omissions, duplicate conflicts, and chain practices can affect measurement. State clustering does not address all dependence across national chains. Income estimates have uncertainties that are not propagated here. The specialist outcome is sparse, and its linear projection yields 80 negative fitted values; these are not interpreted as probabilities, and the nonnegative Poisson sensitivity is reported. Future work should validate menus in a stratified sample, benchmark listing coverage against an establishment registry, and add more suitable control variables and fixed effects for a better causal design.

References

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Visual appendix

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Table 1: From raw rows to the county analysis sample

Processing step	Retained	Removed
Raw rows in the 380K file	380,358	0
Explicit United States address	305,854	74,504
Valid coordinates in the contiguous US bounding box	301,642	4,212
Unique location IDs and coordinates	137,147	164,495
Unique normalized names and coordinates	137,135	12
Observed primary and secondary categories	137,121	14
Eating outlet category rule	135,072	2,049
Unique full resolution county polygon match	135,072	0
Complete county income population and land area	135,072	0
Main analysis counties with at least 30 listings	131,807	3,265

Notes: Source: kwxdata, Kaggle, version 2, released October 1, 2023. Only `380K_US_Restaurants.csv` is used; the companion 200k file is not appended. Counts are rows until deduplication and distinct locations thereafter. Duplicate priority is more category information, a valid rating, a longer address, then original row order. The second duplicate check uses normalized name and coordinates rounded to five decimals. County matching uses full resolution 2021 TIGER polygons, with no snapping of unmatched points. Removed counts are sequential. The main sample includes 131,807 outlets in 862 counties across 48 state clusters.

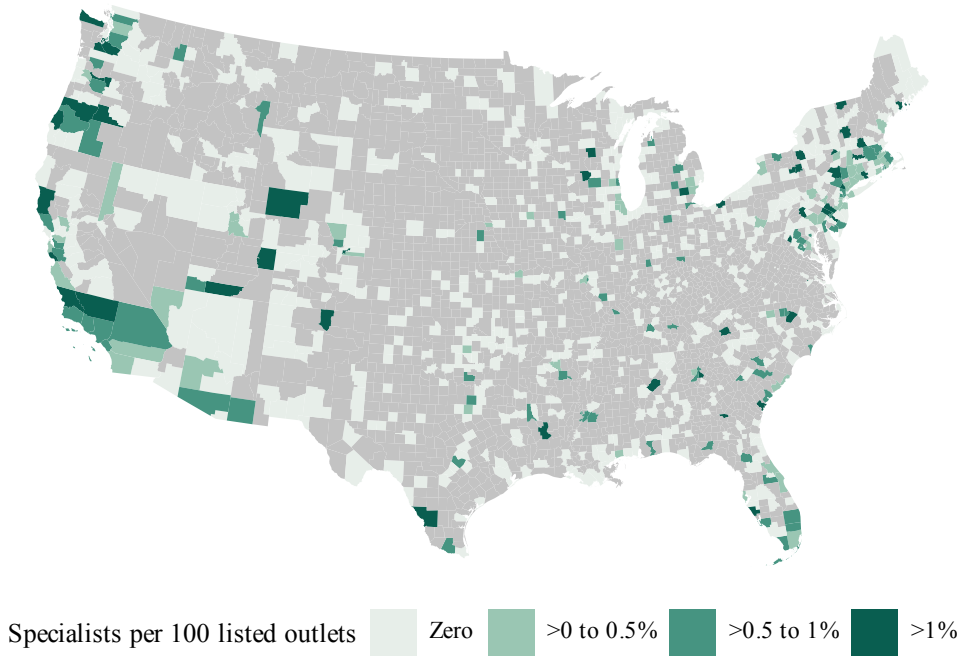


Figure 1: Geographic distribution of primary category vegan or vegetarian specialists

Notes: Share of listed outlets whose primary category is vegan restaurant, vegetarian restaurant, or vegetarian cafe and deli. The denominator is all retained listed eating outlets in each analysis county; county listing counts are at least 30. Light green denotes an observed zero. Gray counties are outside the analysis sample, not observed zeros. The map covers the contiguous United States including DC, uses an Albers equal area projection, and contains 365 specialists among 131,807 listings. Census 2021 cartographic boundaries are simplified at 750 meters for display only; county assignment uses full resolution TIGER polygons. This is not a population representative measure of food access.

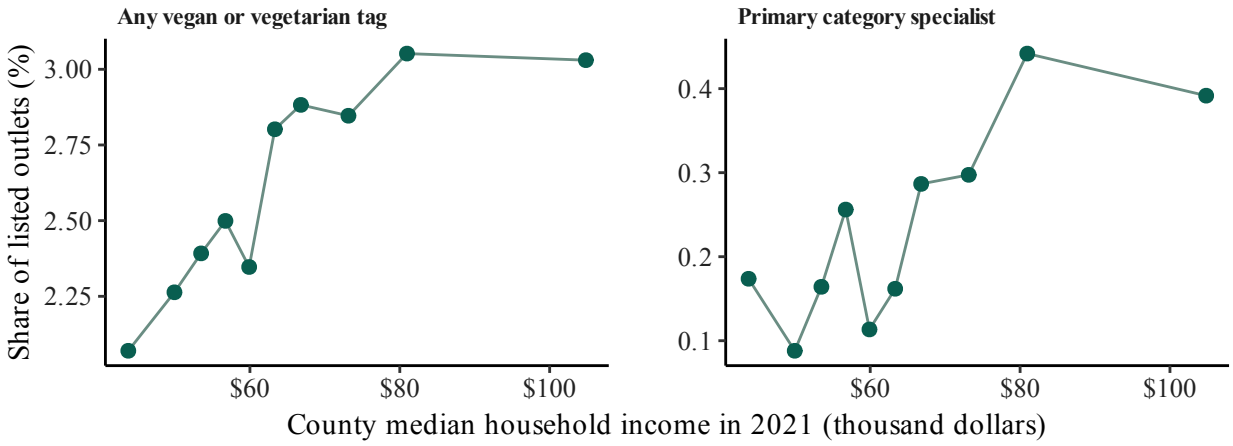


Figure 2: Unadjusted income gradients differ by the definition of plant based dining

Notes: The 862 analysis counties are ranked by 2021 median household income and divided into ten bins with approximately equal numbers of counties. Within each bin, the horizontal axis is mean county income weighted by listing counts and the vertical axis is the pooled tagged count divided by all listed outlets. The two panels have different vertical scales. Lines connect descriptive bin means and do not identify a causal response. Both panels use the same 131,807 outlets. Income is from Census SAIPE 2021; restaurant listings are from Kaggle version 2.

Table 2: Conditional income associations and sensitivity checks

Outcome and specification	$\hat{\beta}$	95% interval	Counties	States
Any vegan or vegetarian tag	0.561	[0.047, 1.075]	862	48
Primary category specialist	0.024	[-0.104, 0.152]	862	48
Any tag excluding Taco Bell	0.710	[0.137, 1.283]	862	48
Any tag without density control	0.842	[0.403, 1.282]	862	48
Any tag equal county weights	0.392	[-0.147, 0.932]	862	48
Any tag at least 50 outlets	0.626	[0.086, 1.167]	746	48
Any tag excluding repeated names	0.579	[0.003, 1.155]	862	48
Specialist equal county weights	0.059	[-0.064, 0.181]	862	48

Notes: $\hat{\beta}$ is the change in the percentage share of tagged listings associated with a doubling of county median household income. Except where stated, regressions weight counties by their relevant listing counts and include log base 2 population per square kilometer and state fixed effects. Each interval uses HC1 state clustered covariance, a cluster count correction, and a t_{G-1} critical value; G is the States column. No significance stars are used. The first two rows use the common sample of 131,807 listings. Excluding Taco Bell removes those outlets from both numerator and denominator; excluding repeated names removes normalized names appearing at least ten times in the cleaned location sample and is not a verified independent restaurant classification. Those two checks keep the original set of eligible counties when a positive denominator remains. The 50 outlet check changes county eligibility.

AI usage declaration

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Academic task	AI used	Tool	Purpose and verification
Research question and hypothesis	No	-	-
Literature review	Yes	ChatGPT (Codex) and Gemini	Summary supporting research and official source documentation. Bibliographic details and the scope of cited claims were checked against primary records.
Data preparation	Yes	ChatGPT (Codex)	Designed category measures, and spatial joins, outcome count bounds, and sequential attrition were checked in R.
Code writing and debugging	Yes	Visual Studio Code	Wrote and debugged the R pipeline and LaTeX source.
Analysis and method choice	Yes	ChatGPT (Codex)	Selected weighted linear projections, state clustered inference, Poisson sensitivity, and alternative definitions. The results are associational and do not identify an income effect.
Results interpretation	No	-	-
Writing and slide preparation	Yes	Overleaf	Overleaf code and review edited the brief, appendix, documentation, and this declaration.